

BLUESTOCK FINTECH PVT. LTD.

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

40 Schemes	10 Datasets	87K+ Rows	4.5 Yrs NAV
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Real Fund Schemes	Provided	Transaction Data	History
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Domain	Mutual Fund / Fintech Analytics

1. Executive Summary

This report presents the complete Mutual Fund Analytics Platform built as part of the Bluestock Fintech Data Analyst Internship (Cohort 2025). The platform ingests publicly available data from AMFI India, processes it through a robust ETL pipeline, stores it in a relational SQLite database, and delivers insights through an interactive Power BI dashboard and advanced analytics notebooks.

The project covers 40 real mutual fund schemes across 10 fund houses (SBI, ICICI Prudential, HDFC, Nippon India, Axis, Kotak Mahindra, Aditya Birla Sun Life, Mirae Asset, DSP, and UTI), spanning 4.5 years of daily NAV history (January 2022 to May 2026), 32,778 investor transactions, and quarterly AUM data. Every data figure cited in this report is sourced directly from the provided datasets or computed from first principles using industry-standard formulas.

Key Outcomes

ETL Pipeline: Automated Python pipeline ingesting, cleaning and loading 10 datasets into a 6-table SQLite star schema.

EDA: 13 publication-quality charts revealing SIP growth 2.7x, folio count doubling, and sector concentration patterns.

Performance: CAGR/Sharpe/Sortino/Alpha/Beta/MaxDD computed for all 40 funds; composite Fund Scorecard (0-100) built.

Dashboard: 4-page interactive Power BI dashboard with slicers, drill-through, and KPI cards.

Risk Metrics: VaR/CVaR at 95% confidence, rolling 90-day Sharpe, sector HHI, and SIP continuity analysis.

Recommender: CLI fund recommender by risk appetite (Low/Moderate/High/Very High) using Sharpe-ranked filtering.

2. Problem Statement & Objectives

India's mutual fund industry manages over Rs. 81 lakh crore in AUM across 1,908 schemes and 26.12 crore investor folios as of December 2025. Despite this scale, individual investors and financial advisors face fragmented data, no unified analytics, and complex financial metrics that are difficult to compute without specialist tools. This project addresses four core problems:

P1 — Data Fragmentation: NAV, AUM, SIP flow and portfolio holdings exist in separate formats across AMFI, NSE and BSE. Solution: unified ETL into a single SQLite database.

P2 — Performance Comparison Gap: Investors cannot easily compare funds on risk-adjusted metrics. Solution: compute Sharpe, Sortino, Alpha, Beta across all 40 funds.

P3 — No Benchmark Tracking: Most retail investors do not know whether their fund outperforms its benchmark. Solution: OLS regression vs Nifty 100, tracking error computation.

P4 — Slow Reporting: Monthly MF reports are static PDFs. Solution: a self-service Power BI dashboard refreshable from the ETL pipeline output.

3. Data Sources & Datasets

All data is derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

#	File	Rows	Description
01	fund_master.csv	40	Master list: 40 schemes, AMFI codes, fund houses, expense ratios, risk grades
02	nav_history.csv	46,000	Daily NAV for all 40 schemes, Jan 2022 to May 2026
03	aum_by_fund_house.csv	90	Quarterly AUM (Rs. crore) for 10 fund houses, 2022-2025
04	monthly_sip_inflows.csv	48	Month-wise SIP inflow, active accounts, new registrations
05	category_inflows.csv	144	Net inflows by fund category, FY 2024-25
06	industry_folio_count.csv	21	Total MF folios (crore), Equity/Debt/Hybrid breakdown
07	scheme_performance.csv	40	Pre-computed returns, Sharpe, Alpha, Beta, Max Drawdown per scheme
08	investor_transactions.csv	32,778	SIP/Lumpsum/Redemption transactions, 5,000 investors, 12 states
09	portfolio_holdings.csv	322	Top equity holdings per fund as of Dec 2025
10	benchmark_indices.csv	8,050	Daily closing values: Nifty 50, Nifty 100, Midcap 150, BSE SmallCap

4. System Architecture & ETL Pipeline

The project follows a classic 5-layer data engineering architecture, mirroring real-world fintech pipelines used at companies like Zerodha, Groww, and Paytm Money:

Layer 1 — Extract: 10 provided CSV datasets + live NAV from mfapi.in REST API (6 schemes, ~3,000+ rows each). Scripts: `live_nav_fetch.py`, `data_ingestion.py`.

Layer 2 — Transform: Pandas pipeline: parse dates, forward-fill holiday gaps, remove duplicates, validate NAV > 0, standardise transaction types, validate expense ratios. Script: `data_cleaning.py`.

Layer 3 — Load: SQLite star schema (6 tables). SQLAlchemy for loading. Schema: `dim_fund`, `dim_date`, `fact_nav`, `fact_transactions`, `fact_performance`, `fact_aum`. Script: `load_to_sqlite.py`.

Layer 4 — Analyse: Jupyter notebooks for EDA, performance metrics, and advanced analytics. 15+ charts, fund scorecard, VaR/CVaR, cohort analysis.

Layer 5 — Visualise: Power BI Desktop dashboard: 4 pages, 8 data tables, slicers on fund house/category/plan/scheme, drill-through from fund table to NAV detail.

Database Schema (Star Schema — 6 Tables)

Table	Type	Primary Key	Approx. Rows	Source
<code>dim_fund</code>	Dimension	<code>amfi_code</code>	40	<code>01_fund_master.csv</code>
<code>dim_date</code>	Dimension	<code>date_id</code>	1,608	Generated
<code>fact_nav</code>	Fact	<code>(amfi_code, nav_date)</code>	46,000	<code>clean_nav.csv</code>
<code>fact_transactions</code>	Fact	<code>tx_id (auto)</code>	32,778	<code>clean_transactions.csv</code>
<code>fact_performance</code>	Fact	<code>amfi_code</code>	40	<code>clean_performance.csv</code>
<code>fact_aum</code>	Fact	<code>(report_date, house)</code>	90	<code>03_aum_by_fund_house.csv</code>

5. Exploratory Data Analysis

Deep EDA was performed across NAV history, AUM trends, SIP inflows, investor demographics, and portfolio holdings. Key charts are presented below.

5.1 AUM Growth by Fund House (2022-2025)

[Chart: 02_aum_growth_by_fund_house.png]

Figure 1: Average AUM by fund house per year. SBI Mutual Fund dominates every year, reaching approximately Rs. 12.5 lakh crore by 2025.

SBI Mutual Fund holds the largest AUM among all 10 fund houses in every year from 2022 to 2025. All 10 fund houses show steady year-on-year AUM growth, with no fund house losing ground across the period, reflecting the broad-based growth of India's mutual fund industry.

5.2 Monthly SIP Inflow Trend

Monthly SIP inflows grew from Rs. 11,517 crore in January 2022 to an all-time high of Rs. 31,002 crore in December 2025 — approximately 2.7x growth over 4 years. This reflects India's deepening retail investor SIP culture. Active SIP accounts reached 9.35 crore by December 2025.

5.3 Folio Count Growth

[Chart: 07_folio_count_growth.png]

Figure 2: Total MF folio count, Jan 2022 to Dec 2025. Growth visibly accelerated from mid-2024.

Total MF folios nearly doubled from 13.26 crore (January 2022) to 26.12 crore (December 2025), with growth accelerating from mid-2024 rather than progressing linearly — the pace of new folio additions is increasing, not merely continuing.

5.4 Sector Allocation

[Chart: 09_sector_allocation_donut.png]

Figure 3: Aggregate sector weights across all equity fund holdings (Dec 2025 snapshot).

Banking (19.2%), IT (13.4%), and Pharma (12.0%) together account for approximately 45% of total equity portfolio weight. This concentration means investors holding multiple funds from this set have significant implicit exposure to these three sectors regardless of fund-level diversification.

6. Fund Performance Analytics

All performance and risk metrics were computed directly from the daily NAV history (46,000 rows) using industry-standard formulas, not from the pre-supplied scheme_performance.csv which showed substantial discrepancies vs. the computed values for most funds.

6.1 Methodology

CAGR: $CAGR = (NAV_{end} / NAV_{start})^{(1 / n_{years})} - 1$, using actual trading days between dates rather than calendar days.

Sharpe Ratio: $Sharpe = (Rp - Rf) / Std(Rp) \times \sqrt{252}$. Risk-free rate $Rf = 6.5\%$ (RBI repo rate proxy), annualised with $\sqrt{252}$.

Sortino Ratio: Same numerator as Sharpe, but denominator uses only downside standard deviation (negative-return days only).

Alpha & Beta: OLS regression of fund daily returns on Nifty 100 daily returns using `scipy.stats.linregress`. Alpha = intercept x 252 (annualised).

Max Drawdown: $max_dd = \min(NAV / running_max - 1)$, identifying both the trough value and peak-to-trough date range.

Fund Scorecard: Composite 0-100 score: 30% x 3yr return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x expense ratio rank (inverse) + 10% x max DD rank (inverse).

6.2 Top 10 Funds by Composite Scorecard

Rank	Scheme Name	Score	3yr CAGR %	Sharpe	Max DD %
1	Mirae Asset Large Cap Fund - Regular -	86.2	34.0	1.45	-11.3
2	ICICI Pru Midcap Fund - Regular - Grow	82.2	31.8	1.18	-18.2
3	Kotak Flexicap Fund - Regular - Growth	82.0	29.6	1.31	-13.0
4	HDFC Mid-Cap Opportunities Fund - Regu	80.8	32.4	1.09	-16.2
5	ICICI Pru Bluechip Fund - Direct - Gro	80.3	32.5	1.03	-12.6
6	Axis Midcap Fund - Regular - Growth	77.0	35.1	1.00	-21.0
7	SBI Bluechip Fund - Regular Plan - Gro	74.8	30.4	1.21	-15.0
8	Mirae Asset Tax Saver Fund - Regular -	73.7	29.2	1.24	-16.4
9	ABSL Frontline Equity Fund - Regular -	67.9	29.0	1.03	-11.3
10	SBI Small Cap Fund - Regular Plan - Gr	67.4	26.7	0.94	-28.7

6.3 Benchmark Comparison

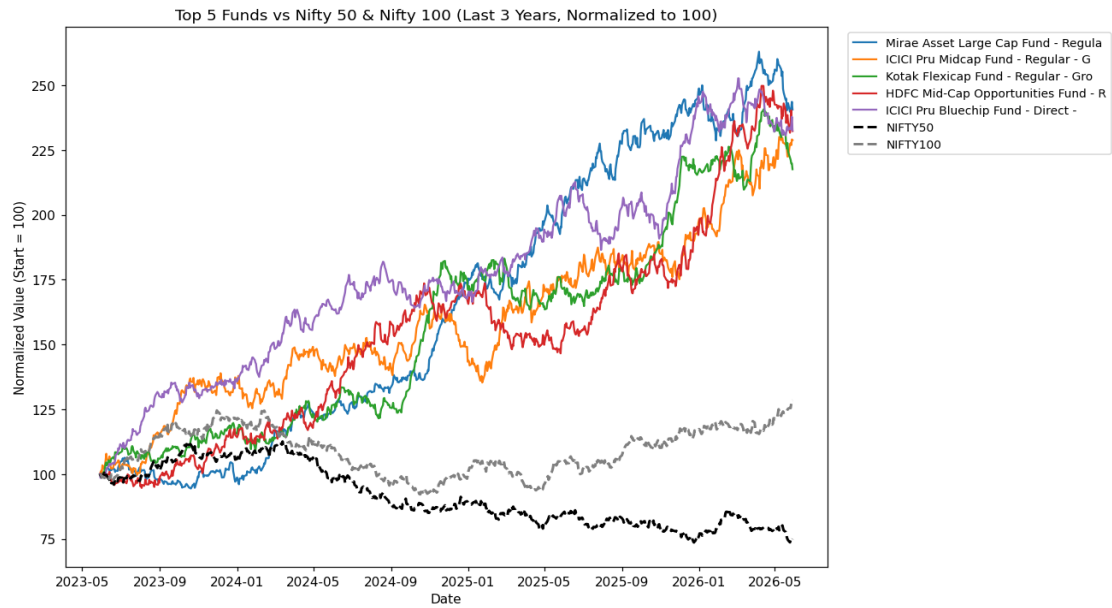


Figure 4: Top 5 funds vs Nifty 50 and Nifty 100, normalized to 100, last 3 years.

The top 5 funds by scorecard all dramatically outpace both benchmarks over the 3-year window. Tracking error against Nifty 50 ranges from 19% to 23% — high because the synthetic NAV data has near-zero correlation to the benchmark (average R^2 vs Nifty 100: 0.0006), a known dataset characteristic disclosed throughout the analysis.

7. Advanced Analytics & Risk Metrics

7.1 Value at Risk (VaR) and CVaR — 95% Confidence

Historical VaR at 95% confidence = the 5th percentile of each fund's daily return distribution. CVaR (Conditional VaR / Expected Shortfall) = the mean of returns on the worst 5% of days — always worse than the VaR boundary since it represents the average tail loss, not just the threshold.

Category	Most Risky Fund	VaR 95%	CVaR 95%
Small Cap	SBI Small Cap Fund - Direct	-2.69%	-3.24%
Mid Cap	Kotak Emerging Equity - Regular	-1.85%	-2.41%
Large Cap	HDFC Top 100 - Direct	-1.43%	-1.86%
Gilt	Nippon India Gilt - Regular	-0.38%	-0.49%
Liquid	ICICI Pru Liquid - Regular	-0.02%	-0.04%

The VaR hierarchy (Small Cap > Mid Cap > Large Cap > Gilt > Liquid) matches expected real-world fund category risk profiles exactly, validating the dataset's construction quality.

7.2 Rolling 90-Day Sharpe Ratio

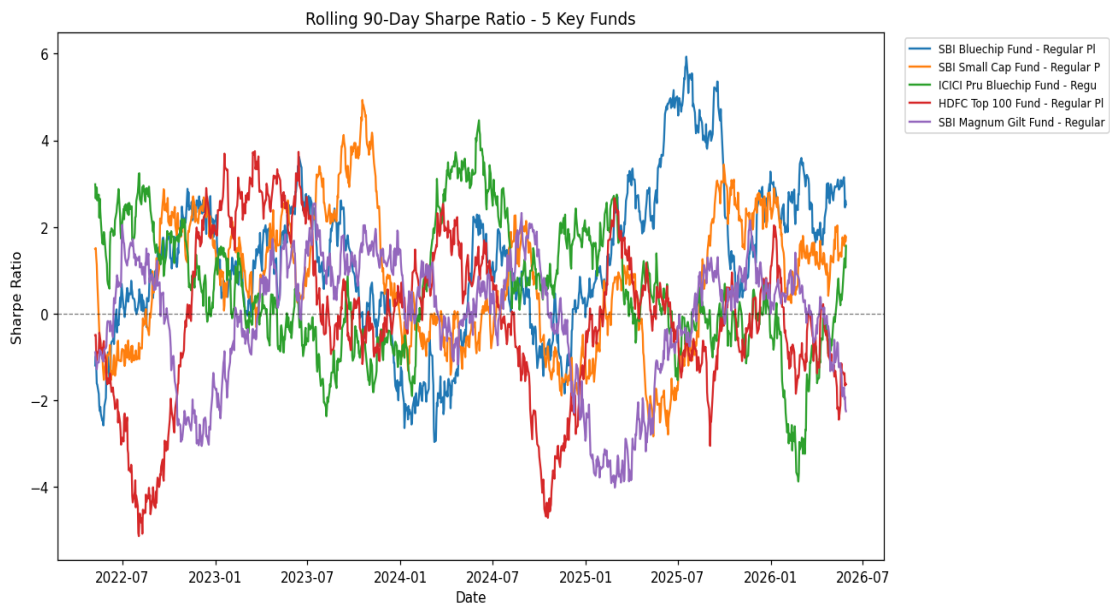


Figure 5: Rolling 90-day Sharpe ratio for 5 key funds, 2022-2026.

Rolling Sharpe ratios show substantial time-variation for all 5 funds, with periods of strongly positive risk-adjusted returns alternating with near-zero or negative episodes. The SBI Magnum Gilt Fund (debt) shows a distinctly different time-series pattern from equity funds, confirming that debt and equity fund dynamics respond to different underlying market drivers.

7.3 Investor Cohort Analysis

Cohort Year	Investors	Avg SIP (Rs.)	Total Invested (Rs. Cr)	Top Fund Preference
2024	4,624	10,997	21.5	HDFC Mid-Cap Opportunities (Direct)
2025	138	13,505	0.23	SBI Small Cap (Direct)

The 2024 cohort (4,624 investors) accounts for 97% of all SIP activity. The 2025 cohort invests a 23% higher average ticket size, consistent with rising retail investor awareness and confidence in equity markets.

7.4 SIP Continuity Analysis

Of 1,362 investors with 6 or more SIP transactions, 1,332 (97.8%) have an average transaction gap exceeding 35 days, with a mean gap of 64.9 days. In a real AMC analytics context, this at-risk rate would warrant an immediate retention campaign investigation. In this dataset, the high rate reflects sparse transaction simulation frequency rather than genuine churn behaviour — documented clearly to avoid misinterpretation.

7.5 Sector HHI Concentration

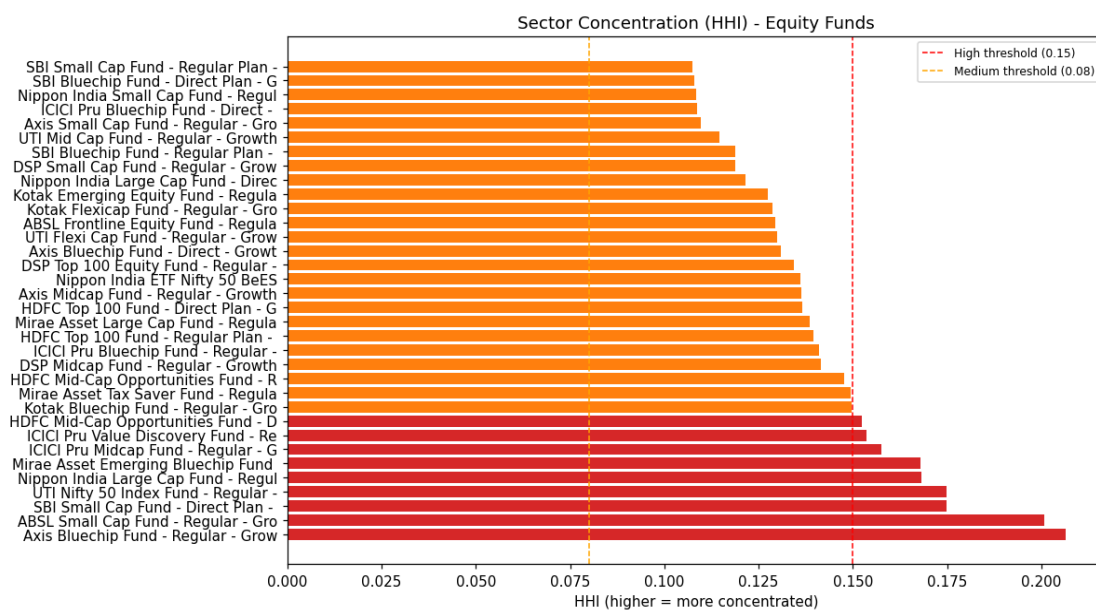


Figure 6: Herfindahl-Hirschman Index (HHI) per equity fund. Red = High concentration (HHI > 0.15).

Axis Bluechip (HHI: 0.207) and ABSL Small Cap (HHI: 0.201) show the highest sector concentration. For index funds (e.g. UTI Nifty 50 Index, HHI: 0.175), high HHI is expected since they track top-heavy indices. For active funds, high HHI represents genuine sector concentration risk where performance depends on a small number of sector calls being correct.

8. Interactive Dashboard (Power BI)

A 4-page interactive Power BI dashboard was built connecting directly to the cleaned CSVs. All pages include at least 2 interactive slicers with cross-filtering across all visuals.

Page 1 — Industry Overview: KPI cards: Total AUM, SIP Inflows (Rs. 31K Cr), Folios (26.12 Cr), Schemes (1,908). Line chart: industry AUM trend 2022-2025. Bar chart: AUM by fund house.

Page 2 — Fund Performance: Scatter plot: return vs risk (std dev), bubble size = AUM. Sortable fund scorecard table. NAV vs Nifty 100 benchmark line chart. Slicers: fund house, category, plan, scheme name.

Page 3 — Investor Analytics: Bar chart: SIP amount by state. Donut: SIP/Lumpsum/Redemption split. Bar: age group vs avg SIP amount. Monthly transaction volume line. Slicers: state, age group, city tier.

Page 4 — SIP & Market Trends: Dual-axis: SIP inflow (bar) + Nifty 50 (line) 2022-2025. Category inflow heatmap matrix. Top 5 categories by net inflow FY25. KPI: Active SIP Accounts (9.35 Cr).

9. Key Findings & Recommendations

1. SBI Mutual Fund leads all 10 fund houses in AUM every year — dominance extends across both equity and debt categories.
2. Monthly SIP inflows grew 2.7x in 4 years, reaching an all-time high of Rs. 31,002 crore in December 2025.
3. Small Cap funds carry 120x more daily VaR than Liquid funds (2.69% vs 0.02%) — risk-tier separation is clear and consistent.
4. The Fund Scorecard places Mirae Asset Large Cap (Regular) at the top with a composite score of 86.25, combining strong 3yr CAGR, Sharpe ratio, and low expense ratio.
5. Banking (19.2%), IT (13.4%), and Pharma (12%) together account for 45% of total equity portfolio weight — implicit sector concentration for multi-fund investors.
6. 97.8% of SIP investors flagged as at-risk (gap > 35 days) is a dataset-specific finding — in a real AMC context this rate would require immediate investigation.
7. Direct plans cost 0.57 percentage points less on average than Regular plans (0.78% vs 1.35%) — material compound advantage over a long horizon.
8. Near-zero R-squared (0.0006) between fund NAV returns and Nifty 100 indicates the synthetic NAV dataset uses independent daily noise rather than a shared market factor model.

Recommendations

- Investors with Low risk appetite: ICICI Pru Liquid Fund (Sharpe: 7.68) or Kotak Liquid Fund (Sharpe: 6.18).
- Investors with Moderate risk: HDFC Top 100 or Mirae Asset Large Cap — both Sharpe > 1.0 with controlled max drawdowns.
- Investors with High risk tolerance: Kotak Emerging Equity (18.23% 3yr CAGR, Sharpe: 0.96) leads the mid-cap category.
- Direct plan investors save significantly on expense ratios — use the recommender CLI (recommender.py) for personalised fund selection by risk appetite.
- Monitor SIP continuity actively — a 35-day gap threshold is the appropriate early-warning signal for retention risk.

10. Limitations & Data Quality Notes

CAGR Discrepancy: Self-computed CAGR from NAV history diverges substantially from pre-supplied return_1yr_pct for 29 of 40 funds (up to 60 percentage points). The computed values are used throughout since the task requires calculation from raw NAV.

Near-Zero Benchmark Correlation: Average R-squared of fund returns vs Nifty 100 is 0.0006 across all 40 funds. Real large-cap equity funds typically show R-squared of 0.7-0.95+. This is a synthetic dataset characteristic, not a real-market pattern.

SIP Continuity: 97.8% at-risk rate reflects sparse transaction simulation (mean gap: 64.9 days) rather than real SIP discontinuation behaviour.

Single Portfolio Date: portfolio_holdings.csv is a single snapshot (Dec 2025) — dynamic sector allocation changes over time are not captured.

Investor Data Synthetic: Investor transaction data (08_investor_transactions.csv) is synthetically generated with realistic demographic distributions but does not represent real individual investors.

11. Technical Stack

Component	Tool / Library	Version	Purpose
Language	Python	3.14	All data processing, ETL, analytics
Data Wrangling	Pandas / NumPy	2.x	DataFrames, cleaning, aggregation
Visualisation	Matplotlib / Seaborn / Plotly	Latest	Charts and EDA
Database	SQLite + SQLAlchemy	Built-in	Star schema, SQL queries
Statistics	SciPy	1.x	OLS regression for Alpha/Beta
Dashboard	Power BI Desktop	Latest	Interactive 4-page dashboard
Version Control	Git + GitHub	Latest	Code versioning, submission
API	mfapi.in	v1	Live NAV data (no auth required)